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Technical Skills Information Extraction From Resumes Using Advanced Natural Language Processing Model With Transfer Learning

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Article

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TECHNICAL SKILLS INFORMATION EXTRACTION FROM RESUMES USING ADVANCED NATURAL LANGUAGE PROCESSING MODEL WITH TRANSFER LEARNING

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ABSTRACT

Matching job seekers with suitable job vacancies in the rapidly evolving job market is challenging. Enhancing the accuracy of job matching requires identifying specific skills and techniques essential for each vacancy. This study proposes an innovative algorithm that utilizes Transfer Learning combined with advanced Natural Language Processing (NLP) models specifically SpaCy, BERT (Bidirectional Encoder Representations from Transformers), and XLNet to automatically detect and extract skills and techniques from job vacancy descriptions. The algorithm initiates by pre-processing job descriptions through tokenization and text cleaning. Next, SpaCy is used for part-of-speech tagging, entity recognition, and syntactic parsing to extract relevant keywords and phrases. These pre-processing steps effectively reduce noise and improve the quality of extracted data. BERT and XLNet, two cutting-edge transformer-based models, are then fine-tuned on a large corpus of job description data. This fine-tuning adapts the models to the specific language and context of job descriptions, enabling them to encode reprocessed descriptions into contextual embeddings. At the algorithm core is the identification of skills and techniques within job descriptions by leveraging these contextual embeddings. A combination of named entity recognition, keyword extraction, and context-based keyword filtering is employed to identify essential skills and techniques for each role. The algorithm's performance is evaluated using precision, recall, and F1-score metrics on labeled job description datasets. Results show the algorithm effectiveness in detecting skills and techniques, providing significant benefits in automating and enhancing the job matching process. By accurately identifying required skills, job seekers can connect with more relevant opportunities, and employers can streamline recruitment. This approach has the potential to improve job market efficiency, leading to more successful placements.

Keywords: Skill Detection, Technique Detection, Job Vacancies, Transfer Learning, SpaCy, BERT (Bidirectional Encoder Representations from Transformers), XLNet, NLP (Natural Language Processing), Named Entity Recognition

1. INTRODUCTION

Companies use data daily to gather information about their customers, aiming to predict future trends, develop new marketing strategies, or recommend items tailored to individual profiles. Often, this data is processed to provide users with more personalized experiences. Out of this need, recommendation systems have emerged. These systems analyse and manage data to suggest items that best fit a user's preferences and profile [1]. Recommendation systems have become a regular part of decision-making whether recommending products based on purchase history or suggesting movies on streaming platforms like Netflix. In a business context, where a company may receive a large number of resumes and must process extensive information on each candidate, managing and analysing this data to select the ideal candidate within a reasonable timeframe can be challenging, often beyond human capacity. Since personnel selection is frequently a time-sensitive task, this tool was developed to expedite the process by ranking resumes according to their relevance to a specific job offer.

In this system, a resume recommendation system is implemented to match applicants with various job roles at a technology company. As in any hiring process, some candidates may possess certain required skills but may not be fully suited to the role, perhaps due to a lack of proficiency in specific programming languages or limited knowledge of certain topics. For such cases, a second recommendation system is available, suggesting training courses to help candidates acquire the necessary skills to qualify for similar roles in the future. It's important to note that this recommendation system does not decide which candidate should be hired but instead accelerates the hiring process using Artificial Intelligence (AI) techniques. As technology has become integral to hiring, the job market has become more dynamic. In Paul and Lal (2021), the authors reference a fourth industrial revolution, highlighting the significant impact of new technologies on the workforce. Similarly, Pinzone et al. (2017) discuss the importance of Industry 4.0, with technologies like the Internet of Things, cloud computing, and Big Data driving changes in the market.

One approach to monitoring changes in the job market is identifying essential skills listed in job vacancies. By determining the typical skills required in a particular field, trends over time can be observed. Text mining is an effective tool for identifying these skills. In Zanine and Dhawan (2015), the authors define text mining as a statistical and computational approach to text analysis, involving various techniques that work in combination to address specific problems. A key component of text mining is Natural Language Processing (NLP). According to Liddy (2001), NLP uses computational methods to understand human language for tasks like

text classification, sentiment analysis, language translation, spam detection, and named entity recognition.

This study aims to create an NLP model for extracting technical skills from resumes to automate aspects of the hiring process. Three specific objectives guide this work:

- Identify suitable data pre-processing techniques;
- Investigate machine learning models for extracting skills from job descriptions;
- Select and apply the most effective technique.

The primary contribution of this research is the strategic selection of Named Entity Recognition (NER) as the primary method for skill extraction. NER is valuable in analysing text, as it classifies entities and captures the relationships and context between words, making it well-suited for identifying skills in job descriptions. This choice was motivated by NER's capability to understand semantic connections within text, forming a robust framework for our goal. By implementing NER and Transfer Learning with pre-trained models such as BERT, XLNet, and SpaCy, we sought to improve the accuracy and efficiency of skill extraction in data science roles. This approach contributes to NLP advancements and addresses the job market's evolving needs. The contributions of this research include not only the careful selection of the NER technique but also the exploration and assessment of advanced models to optimize skill extraction. These contributions collectively enhance our understanding of effective methods for extracting technical skills from job descriptions, particularly in the fast-paced field of data science.

2. LITERATURE REVIEW

In their study, Employee Recruitment Recommendation Using Profile Matching and Naive Bayes, Bachtiar, Pradana, and Yudiari (2019) discuss the increasing complexity of securing a job within a company, not only due to the high qualifications required but also because of the sheer number of applicants for each position. This influx poses a challenge for companies that conduct the selection process manually, often resulting in prolonged timelines. Korkmaz (2019), in his research on Personnel Selection Method Based on the Topsis Multi-Criteria Decision-Making Method, emphasizes the critical importance of having competent management teams, noting that even a successful boss may struggle without a well-selected team. Rabcan, Vaclavkova, and Blasko (2017) conducted a study utilizing fuzzy decision trees and the C4.5 algorithm for selecting optimal candidates within a security company. By using training and testing sets, they processed data through fuzzy decision trees to classify candidates according to the company's specific criteria. Meanwhile, Cárdenas and Hernández (2020),

through analysis of 35 interviews with recruitment professionals from various key industries in Chile, found that while overtly discriminatory criteria have been publicly delegitimized, such biases persist in practice. Though professionals generally do not admit to discriminatory treatment, they may acknowledge it when directly questioned, often negotiating and softening their positions. Fernández-García (2019) suggests two types of measures to mitigate discrimination in hiring processes: preventive measures and reparative measures. Within both categories, the Anonymous Curriculum Vitae (ACV) is identified as a viable strategy to counteract bias in hiring. However, Fernandez-Garcia also notes the complexity and high cost associated with implementing ACV, as it requires advanced technological tools and robust legal guidance to comply with each country specific legal frameworks. Hmoud and Laszlo (2019) developed an intelligent text processing technique that applies text mining to analyze sentiment in a candidate's written statements. Their system automatically classifies sentiments as positive or negative within unstructured text, aiming to predict job behavior, performance, and potential resignation likelihood. Using a data mining system with 29 rules, their goal was to help organizations select and retain suitable talent for specific roles. Significant developments in Natural Language Processing (NLP) have emerged since Vaswani et al. (2017) introduced the Transformer architecture, which forgoes the recurrent structure in favour of an attention mechanism. This mechanism calculates dependencies between each term's inputs and outputs independently, enabling parallel processing, performance gains, and the handling of large databases. This architecture quickly became foundational, particularly within NLP and, more recently, computer vision. Wolf et al. (2020) describe the Hugging Face Transformers library, comprising three key components: the Tokenizer, which converts raw text into tokens, the Transformer model, which converts these tokens into contextual embeddings, and the Head layer, which uses embeddings to generate task-specific predictions. The Transformer architecture paved the way for new language models. For instance, Devlin et al. (2018) introduced BERT (Bidirectional Encoder Representations from Transformers), a model designed to pre-train on bidirectional representations of unlabeled text. BERT can be fine-tuned with an output layer to address specific tasks, meaning it does not require modifications to the pre-trained model. BERT's pre-training utilized documents from English Wikipedia and BooksCorpus, amounting to approximately 3,300 million words. Through fine-tuning, it achieved notable performance across 11 NLP tasks, including question-answering and text prediction. BERT's success has led to numerous variations aimed at improving performance or efficiency. Chaima et al. (2022) presented the RoBERTa model, which modified BERTs training approach by training longer on more extensive datasets, removing

the next-sentence prediction, using larger sequences, and dynamically adjusting the masking pattern. These changes, along with a new training database, enhanced RoBERTa's performance, surpassing state-of-the-art models on four out of nine GLUE tasks and outperforming BERT across all tested tasks, illustrating the impact of careful model design and data quantity on performance. DistilBERT, developed by Sanh et al. (2019), is another BERT variant, designed as a smaller, faster model with a lower computational cost. DistilBERT maintained 97% of BERT's language comprehension while reducing model size by 40% and inference time by 60%, making it more efficient without compromising output quality. The term "DistilBERT" reflects "Knowledge Distillation," a compression technique where a smaller model is trained to replicate the behaviour of a larger model. Another influential Transformer-based model is the Generative Pre-trained Transformer (GPT), with its latest version being GPT-3. According to Floridi and Chiriatti (2020), GPT-3 is designed to generate sequences of words or data from an input, relying on unsupervised training on vast datasets. Brown et al. (2020) explain that GPT-3 was trained with 175 billion parameters ten times larger than any previous model and can be applied directly to tasks without additional fine-tuning, achieving strong results with minimal task-specific input. Consequently, GPT-3 has broad applications, including translation, summarization, grammatical correction, question-answering, and chatbots (Floridi and Chiriatti, 2020).

3. METHODOLOGY

As mentioned earlier, this study aims to select the most effective technique for extracting skills from job vacancies, with a specific focus on technical skills within the web development field. The goal is to identify both established skills and emerging skills that may develop in this field over time. This research is structured into three main stages, as outlined in Figure 1 below, which illustrates the methodology to be employed.

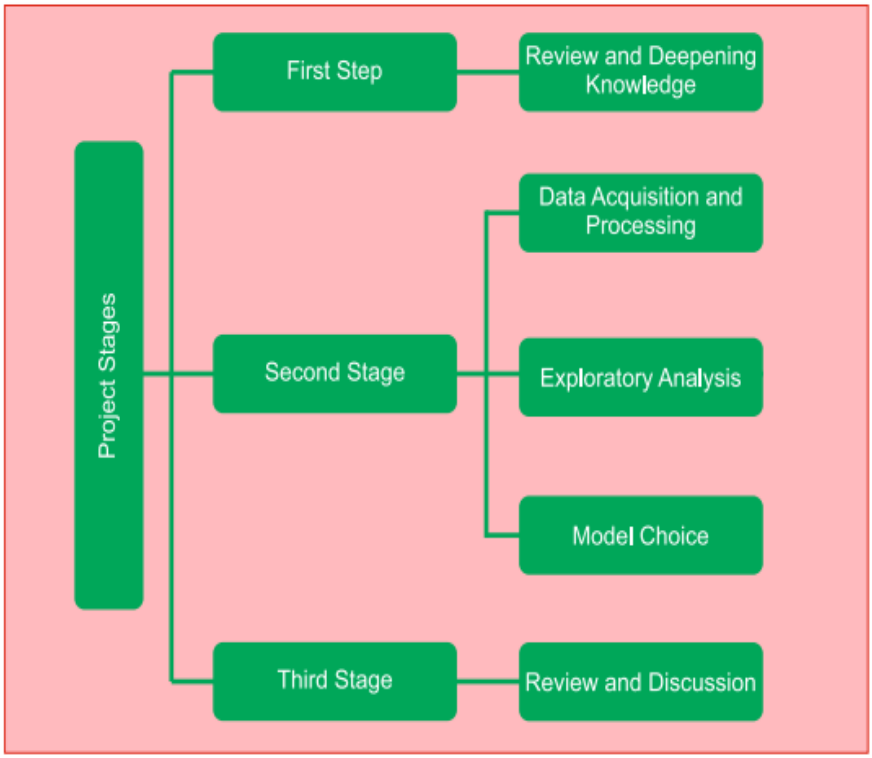


Figure 1. Work Stages

4.1. REVIEW AND DEEPENING KNOWLEDGE

In the first stage, a comprehensive review of the available techniques for extracting skills from text was conducted to deepen knowledge in this area. In the second stage, which is explained in greater detail in the following section, data acquisition and processing, exploratory data analysis, and the selection and creation of the skills extraction model will be performed. In the final stage, the results will be presented, along with discussions and suggestions for future research.

4.2. DATA ACQUISITION

The dataset used in this study was created by collecting job listings from the Indeed website, focusing on native English-speaking countries, specifically Australia, Canada, the United States, Great Britain, and Ireland. To create the initial dataset, job vacancies in Data Science were searched using the keyword “Data Science” on Indeed, targeting the five largest cities by population in each of the selected countries. For automated collection, a web scraper was developed using the Selenium library. The data were saved in country-specific folders in Excel format (.XLSX), resulting in an initial dataset of 6,325 entries, as shown in Figure 2.

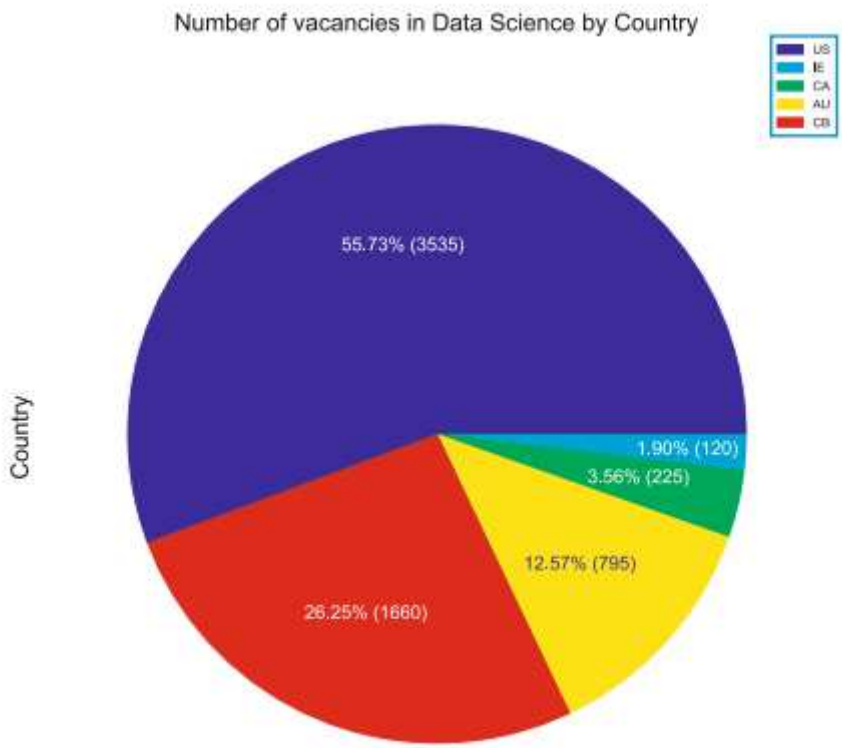


Figure 2. Distribution of Job Vacancies

4.3. DATA PRE PROCESSING

As previously described, the data pre-processing stage aims to refine the data by converting it from raw to prepared form. This essential phase precedes model training, ensuring that the data fed into the model is reliable and ready for analysis.

4.3.1. Feature selection

In text analysis, each word within the text is treated as a feature. In this study, the feature selection process will be divided into two stages: first, selecting which columns from the dataset will be used or discarded, and second, selecting and cleaning the data within the job descriptions.

4.3.1.1. Data selection

The first step in cleaning the dataset involves selecting the features to be used and discarding the others. The *description* column contains the text of each job vacancy, while the *job title* column provides information about the title of the vacancy. The *job title* column will be used to ensure that the listed jobs match the Data Science field. Thus, two columns *description* and *job title* are retained, while the remaining columns are discarded. As a final step in data selection, the job titles retrieved by the web scraper were compared with the search term used on the Indeed job site to confirm that the job listings in the dataset are relevant to the field of interest. Titles were compared using cosine similarity, which returns a decimal

value between 0 and 1 indicating the degree of similarity between two vectors; the higher the value, the closer the similarity between the texts. For this task, a filter was applied to select the 40 job listings with the highest similarity between the job title and the search term “Data Science.” All 40 selected listings had a similarity score of 1, meaning the job title exactly matched the search term.

4.3.1.2. Text cleaning

The data collected by the web scraper was in HTML format, so the first step in cleaning the text was to remove the HTML tags. After analysing the grammatical structure of job listings, it was observed that the majority of relevant skills are represented as nouns. Prior to applying text-cleaning techniques, such as removing stopwords, a new column called *nouns* was created in the dataset as a list, containing all phrases with nouns for each job listing. A copy of the *description* column was also made and preserved without the removal of irrelevant words. As mentioned earlier, removing stopwords is essential when working with textual data, as it eliminates words that do not contribute meaningful information. To accomplish this, the SpaCy library was used for stopword removal.

4.4. EXPLORATORY ANALYSIS

Exploratory analysis is the initial phase of data analysis, aimed at summarizing the characteristics of the dataset. For this exploratory analysis, the *description* column was utilized, retaining the stopwords to preserve the integrity of the text and provide an accurate representation of the data.

4.4.1. Character and word count

Figures 3 and 4 display the character and word counts, respectively. It can be inferred that the character count for each job vacancy ranges from 200 to 7,200, with most documents containing between 2,000 and 3,000 characters. Regarding word count, the number of words varies between 20 and 900, with the most common range being 200 to 400 words.

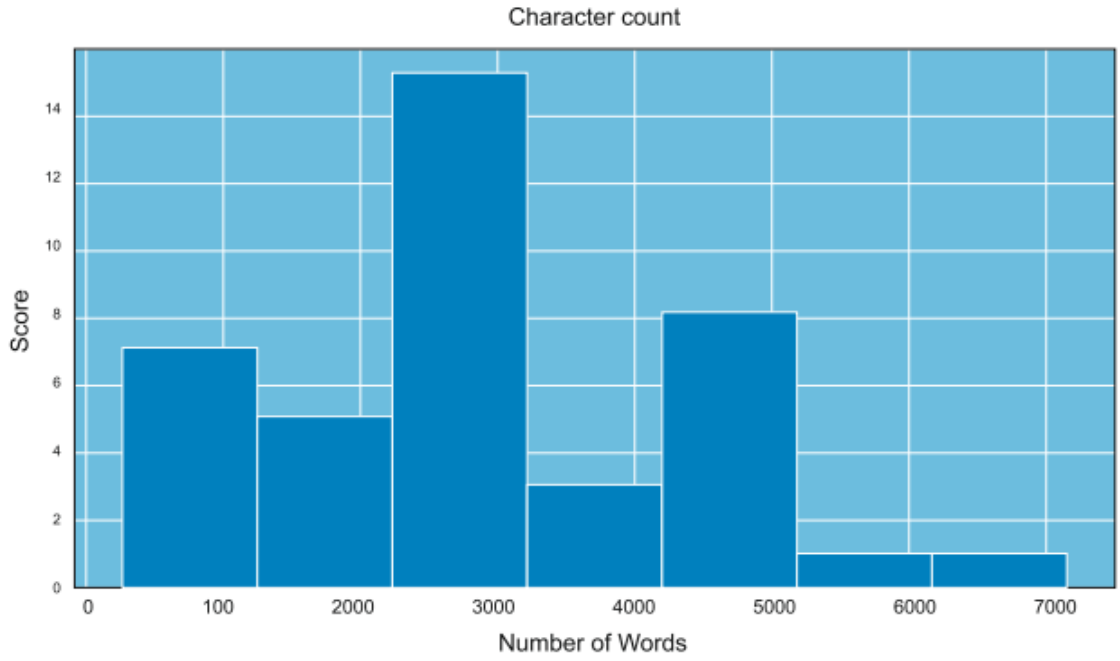


Figure 3. Word Count

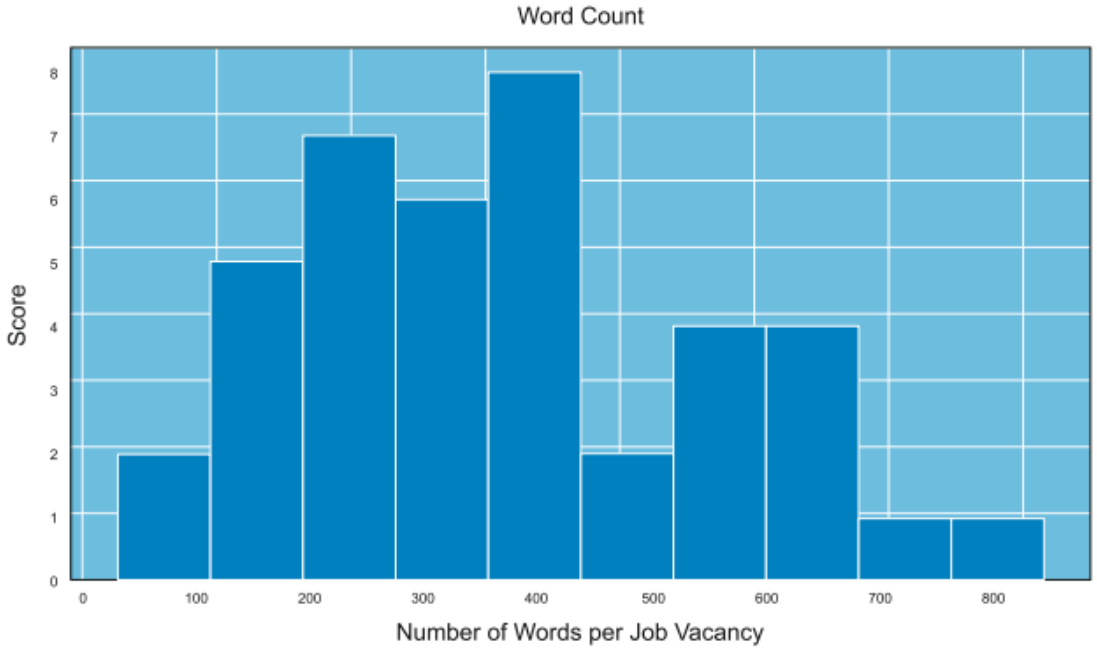


Figure 4. Word Count

4.4.2. Word Length

The average word length ranges from 6.6 to 8 characters, with 6.8 and 7.2 characters being the most common lengths, as shown in Figure 5.

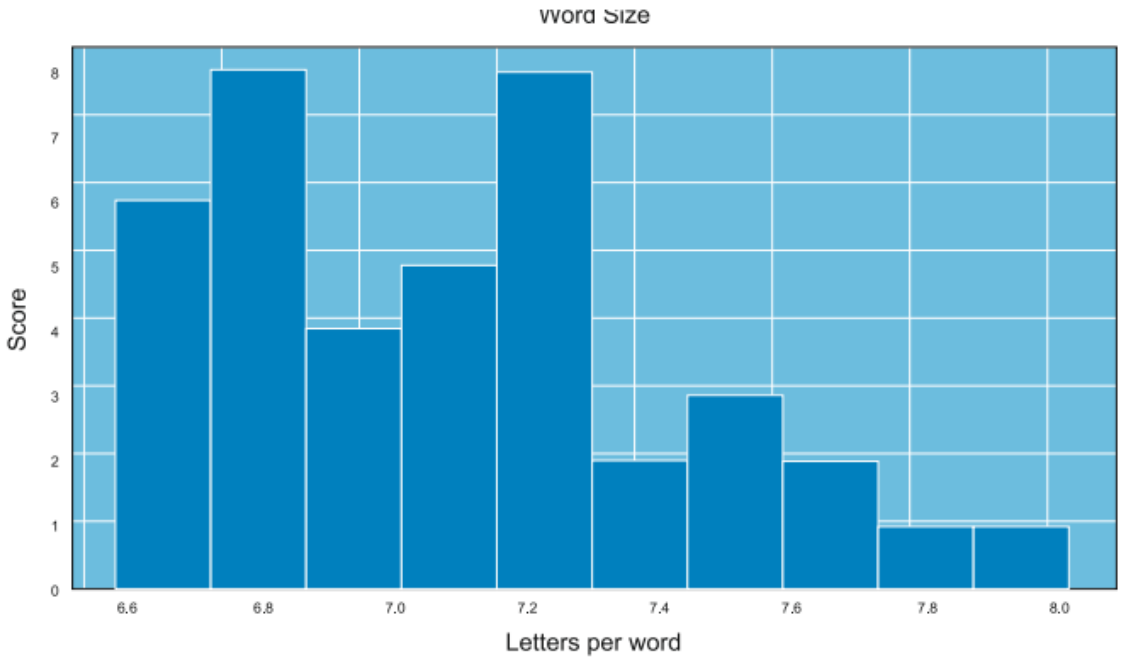


Figure 5. Word Size

4.4.3. Common Words

Figure 6 shows that "data" is the most common word in the corpus, appearing nearly four hundred times, followed by "work" and "experience." Additionally, the words "learn" and "machine" are among the 30 most frequently occurring terms, suggesting that Machine Learning may be one of the most sought-after skills in this field.

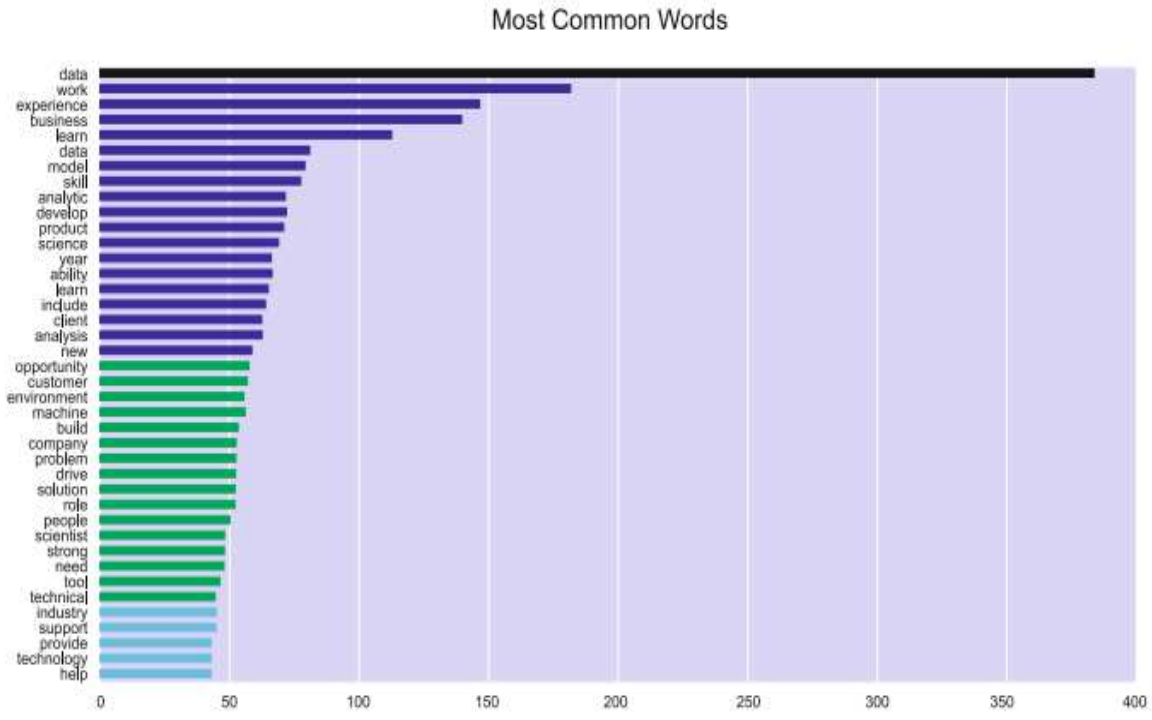


Figure 6. Common Words

4.4.4. Words in Pair

Figure 7 illustrates the pairwise word counts (bigrams). The results indicate that "Data Science" is the most frequently occurring pair of words, followed by "Data Scientist" and "Machine Learning." This reinforces the idea that Machine Learning is likely one of the most sought-after skills in job vacancies within the data science field.

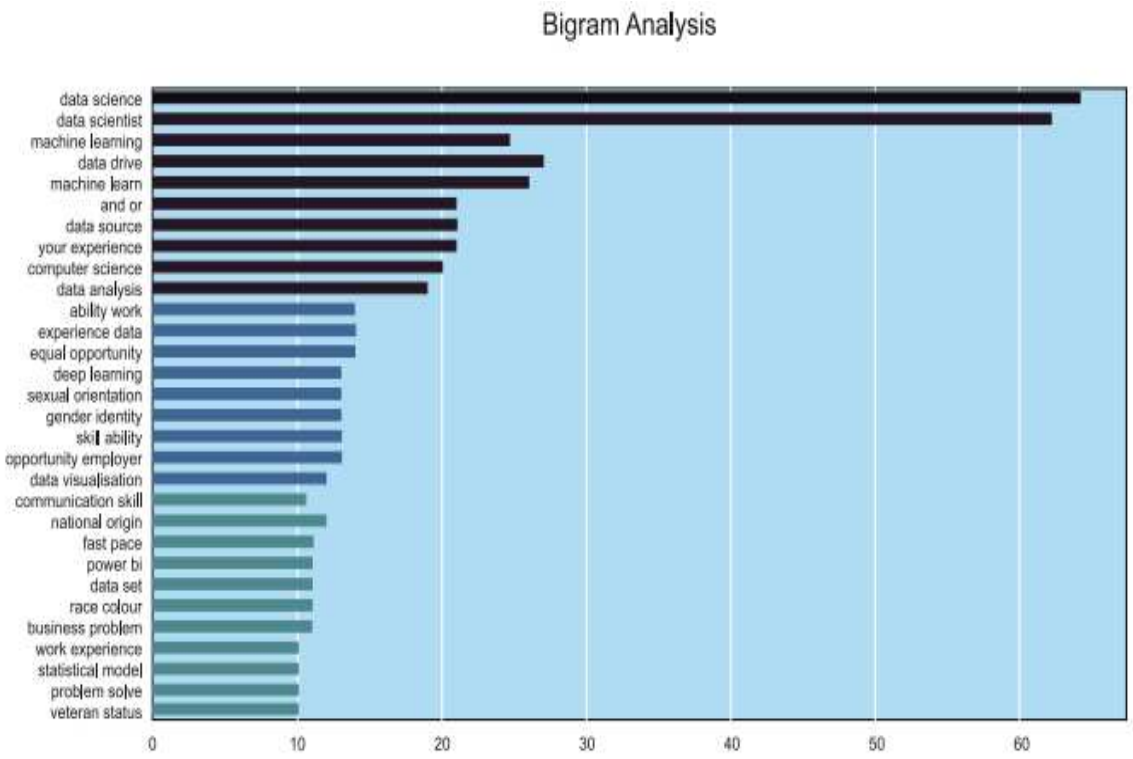


Figure 7. Words in Pair

4.5. CHOICE OF SKILL EXTRACTION TECHNIQUE

Three techniques have been identified as potential candidates for skill extraction: TF-IDF, Word Embeddings, and Named Entity Recognition (NER). In the following subsections, each algorithm will be briefly explained regarding its usefulness for skill extraction. Subsequently, the advantages and disadvantages of each technique will be discussed, leading to the selection of one technique for model creation.

4.5.1. Term Frequency - Inverse Document Frequency (TF-IDF)

As previously mentioned, TF-IDF is an algorithm that quantifies words within a collection of documents. The weights of each word are calculated based on its significance to both the individual document and the overall corpus: the more frequently a word appears in a document, the higher its importance, and the less frequently it appears in the corpus, the greater its importance becomes. This makes TF-IDF particularly useful for identifying skills in job

postings, as it is an unsupervised algorithm that does not require pre-training, enabling it to detect emerging skills. However, one challenge that TF-IDF may encounter in detecting skills in job vacancies is that, in many postings, skills are mentioned only once. This can diminish their weight compared to more frequently occurring words that are not skills, such as "experience" or the job title "Data Scientist."

Unsupervised Learning: TF-IDF is an unsupervised algorithm that does not require pre-training. This characteristic is particularly valuable in the context of job vacancy descriptions, where skills may emerge or evolve over time, making an unsupervised approach highly effective.

Quantifying Importance: TF-IDF quantifies the importance of words within a set of documents, making it particularly useful for identifying emerging skills. Terms that appear frequently in a specific job vacancy but less commonly across all vacancies may indicate sought-after skills.

4.5.2. Noun Classifier

To extract skills from text through classification, the first step is to determine the focus of the classification. The most logical options are to classify paragraphs or sentences that contain skills. The first option, classifying paragraphs, may not be viable since a single paragraph can contain multiple skills or other nouns, making labeling challenging. In contrast, sentence classification is more concrete, as it is rare for two nouns to appear in the same sentence. Below is an example of a possible initial creation process with classifiers, as illustrated in Figure 8:

- Data pre-processing
- Extract sentences containing nouns
- Map sentences into those with skills and those without skills
- Select and train a classifier to predict sentences containing skills; a neural network utilizing Long Short-Term Memory (LSTM) layers could be a strong candidate
- Train the chosen model with the mapped data
- Extract the nouns (skills) from new data based on the sentences predicted to contain skills.

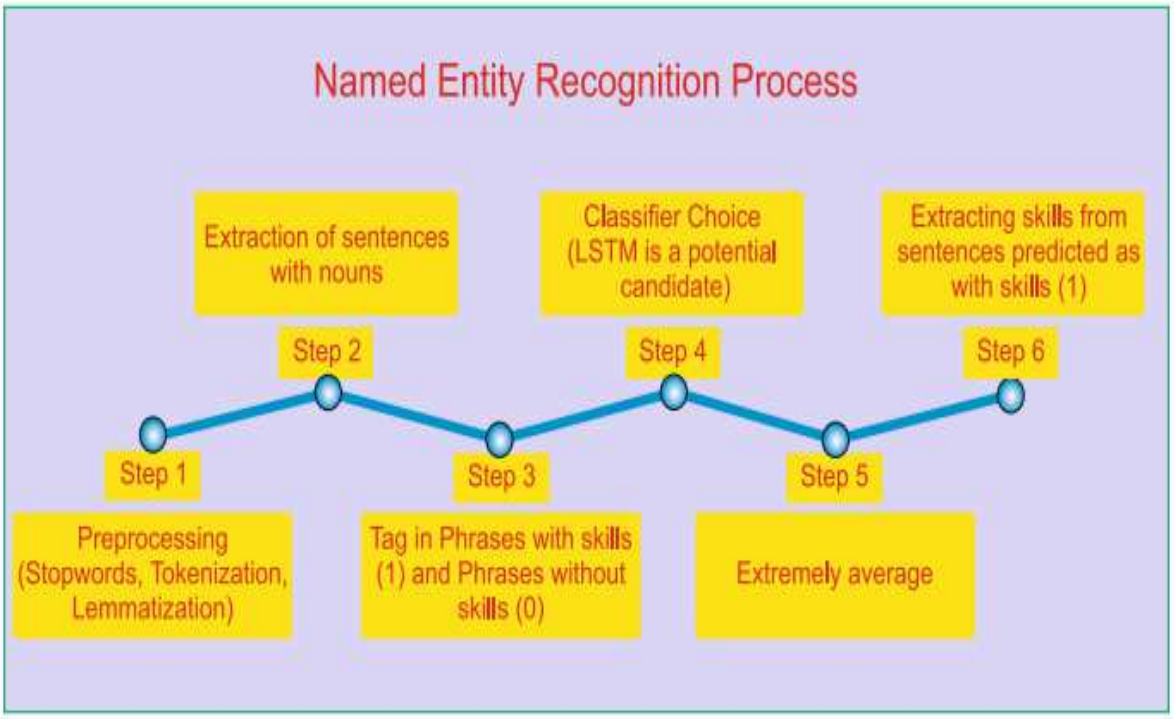


Figure 8. Skills Classification Process

At first glance, classifier extraction appears promising for detecting skills in job vacancies. However, a significant factor that may influence the results is that the classification algorithm does not consider the syntax of the text. It only selects nouns from sentences and classifies them as skills or non-skills.

4.5.3. Named Entity Recognition

Named Entity Recognition (NER) plays a crucial role in enabling machines to understand text written by humans. By classifying the entities within unstructured text, important information such as skills in a job opening can be identified. The algorithm utilizes Word Embeddings as input and employs an artificial neural network, typically a Convolutional Neural Network (CNN) or Long Short-Term Memory (LSTM) network, to comprehend the relationships between words, allowing it to predict whether a word belongs to a specific entity. As mentioned earlier, NER leverages syntax to uncover the meaning behind word relationships, phrases, and texts. Therefore, removing irrelevant words may not be beneficial, as these words can help the model better understand the syntax of the text. A contemporary technique that has shown promising results in this area is Transfer Learning. This involves using a pre-trained model to recognize common entities such as people, dates, and organizations, followed by fine-tuning to adapt it to a specific task, such as recognizing skills in job vacancies.

The entity recognition process is as follows, with Figure 9 illustrating the procedure:

1. Choose a pre-trained model.
2. Create a new entity that will be trained.
3. Label the data, indicating which words belong to the new entity.
4. Perform fine-tuning to train the model to recognize skills.
5. Extract entities (skills) from new job vacancies.

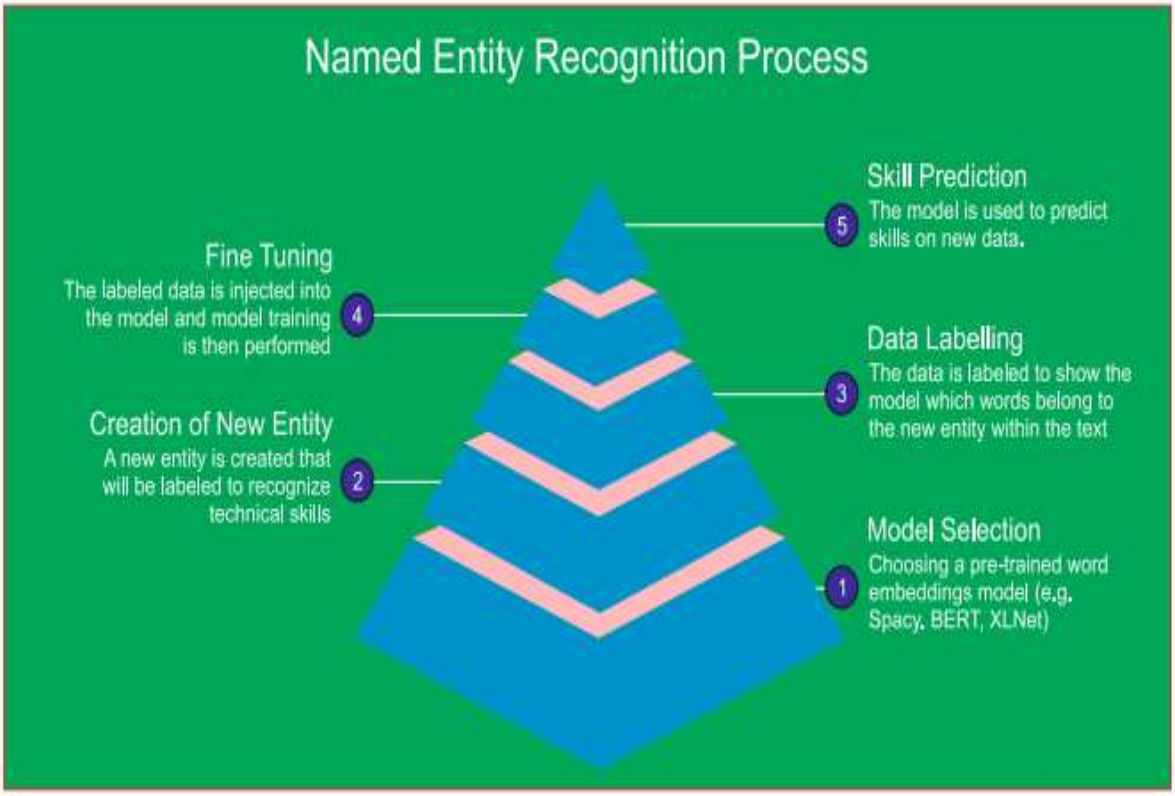


Figure 9. Named Entity Recognition (NER) Process

NER presents a promising approach for extracting skills by leveraging semantics to define entities. However, there are some drawbacks to consider. One significant challenge is the computational power required to execute the algorithm effectively. Additionally, customization is necessary for fine-tuning, as each algorithm has unique characteristics that must be addressed for optimal performance.

Entity Classification: NER is specifically designed to classify entities within unstructured text. In the context of this work, where the objective is to identify technical skills, NER is well-suited for recognizing and categorizing specific entities (skills) mentioned in job vacancies.

Context Understanding: NER models, particularly those based on architectures like BERT or XLNet, excel at understanding the contextual relationships between words. This capability is beneficial for capturing the nuanced context in which skills are mentioned.

4.5.4. Word Embeddings

Semantic Relationships: Word embeddings capture the semantic relationships between words, which is crucial for understanding the context in which skills are mentioned in job descriptions. For example, they can recognize that "machine learning" and "deep learning" are related concepts.

Word Context: Word embeddings, such as those generated by Word2Vec or GloVe, enable the model to understand the context in which words appear. This is particularly beneficial in job descriptions, where skills may be described in various contexts, facilitating accurate extraction.

4.5.5. Choice of Technique

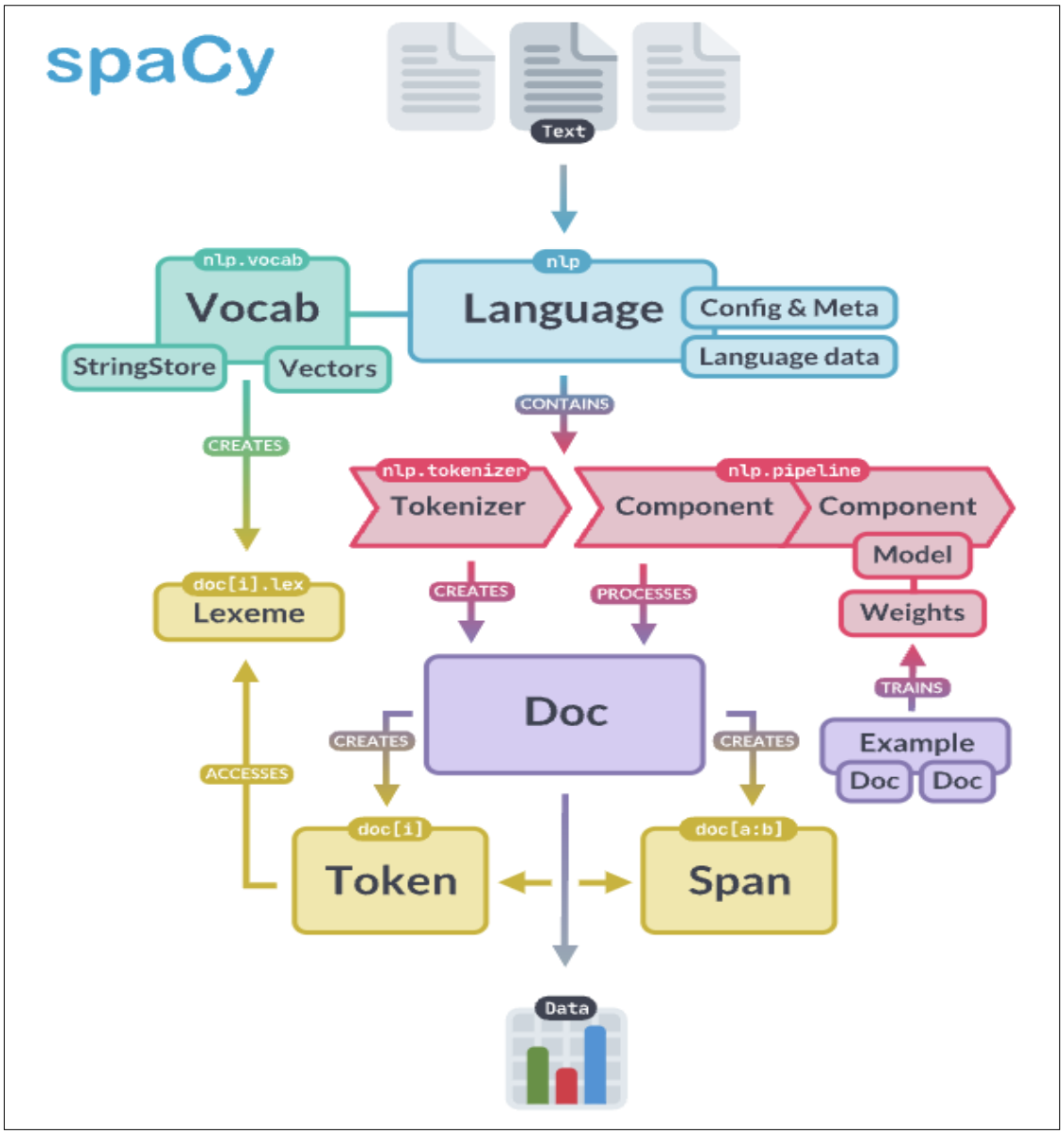
After a careful evaluation of the potential models discussed in the previous sections, Named Entity Recognition (NER) emerged as the most promising choice for extracting skills from job vacancies. This decision is based on NERs utilization of robust pre-trained models, which represent the current state of the art, allowing it to understand the relationships between words and leverage textual syntax to accurately recognize entities.

4.6. MODELS

With data pre-processing completed and the extraction technique selected, the dataset was partitioned into training and validation sets, following a 75% to 25% ratio. These proportions were chosen as they fall within the commonly accepted range in machine learning, which typically spans from 70%-30% to 80%-20% for training and validation data, respectively. The current state of the art in Named Entity Recognition (NER) involves using pre-trained models trained on millions of data points for generalized tasks, followed by fine-tuning these models for specific applications. This approach is known as Transfer Learning. For this work, three pre-trained models were selected for Transfer Learning: Spacy, BERT, and XLNet. The choice of these NLP models-Spacy, BERT, and XLNet was made after careful consideration of their capabilities and their suitability for the task of extracting technical skills from job vacancies. Below are detailed explanations and accompanying diagrams for each model:

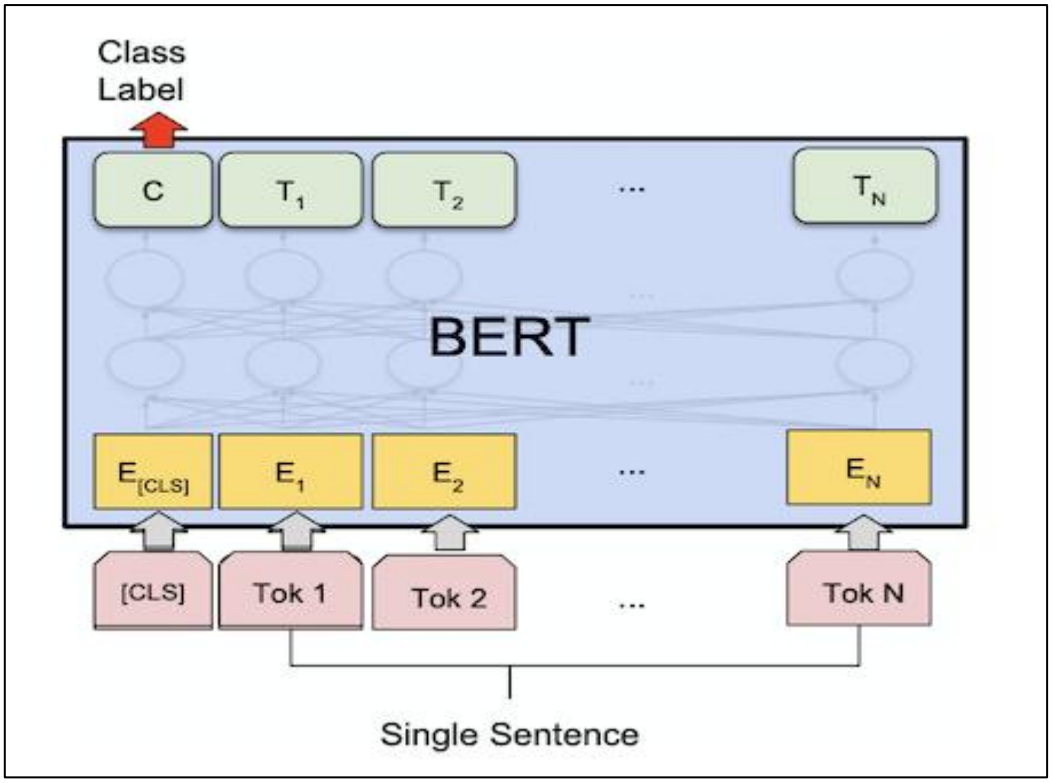
4.6.1. spaCy Model:

spaCy was chosen primarily for its robust Named Entity Recognition (NER) capabilities. The NER model in spaCy employs Word Embeddings and a Convolutional Neural Network (CNN) to recognize entities within a given text. In the context of this work, where the objective is to extract technical skills, spaCy ability to identify and classify specific entities within unstructured text makes it a suitable candidate.



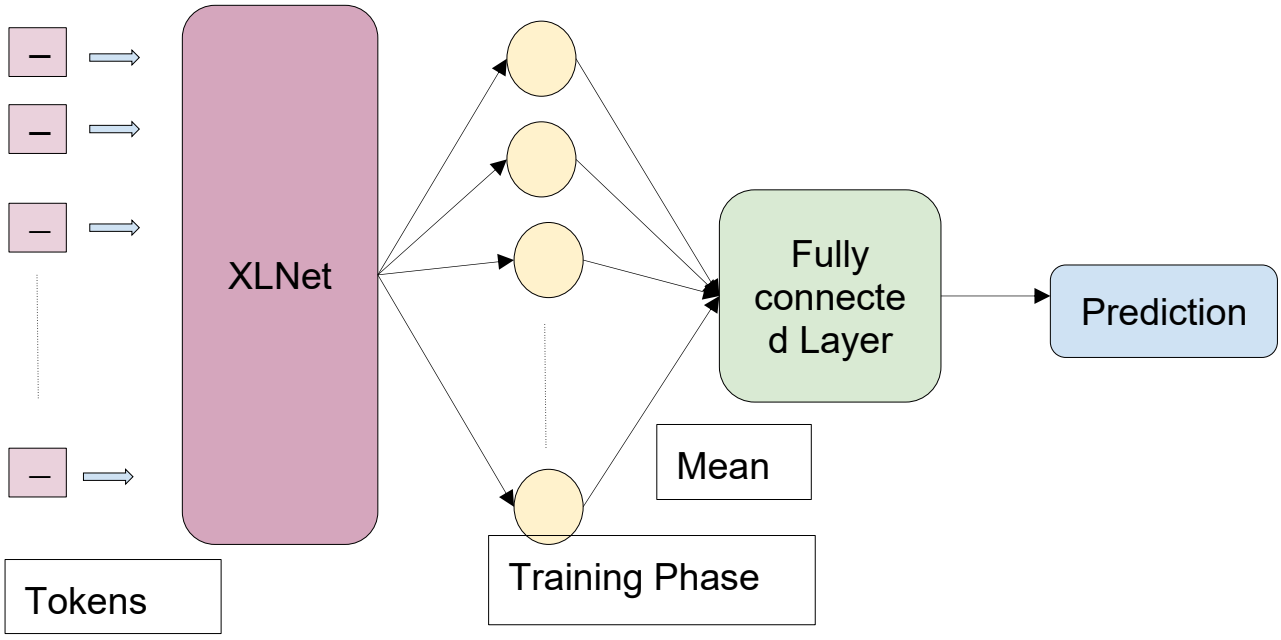
4.6.2. BERT (Bidirectional Encoder Representations from Transformers):

BERT was selected for its bidirectional learning and attention mechanisms, which enable it to understand the contextual relationships between words in a sentence. The model pre-training on extensive datasets, followed by fine-tuning, makes it particularly effective for tasks that require capturing nuanced context, such as identifying technical skills in job vacancies.



4.6.3. **XLNet (Transformer-XL):**

XLNet, an extension of BERT architecture, introduces Permutation Language Modeling. This model retains the bidirectional capabilities of BERT while enhancing its ability to predict words in a sentence by permuting the word order. The combination of bidirectional learning and this permutation approach makes XLNet a powerful model for tasks that require a nuanced understanding of word context.



4.6.4. Why These Models:

The selection of these models is based on their proven success in Natural Language Processing (NLP) tasks. spaCy NER model excels at recognizing entities, aligning well with the objective of identifying technical skills. Both BERT and XLNet, being transformer-based models, are adept at capturing complex contextual relationships, which is essential for understanding job vacancy descriptions where skills are often mentioned in intricate contexts. The decision to utilize these models was driven by their capabilities to handle the nuances of unstructured text and extract relevant information effectively. spaCy is a relatively new library in the NLP domain that offers tools and solutions for various natural language processing tasks. Its NER model employs Word Embeddings and a Convolutional Neural Network (CNN) for entity recognition and has demonstrated impressive results. BERT, developed by Google engineers, utilizes bidirectional learning and encoding techniques to tackle natural language processing challenges through a transformer architecture. BERT's learning approach involves masking certain words during training to predict the correct word. In this work, the RoBERTa version is used, which is a more robust variation of BERT. XLNet is an autoregressive model that also employs bidirectional techniques. It utilizes a method known as Permutation Language Modeling, which allows it to swap words in a sentence to predict one of the words, rather than hiding one and predicting it, as in BERT. All three models were trained for performance comparison, and the results of each model will be presented in the following chapter for the task outlined in this work. To create the new entity and label the data, specific considerations were made to define what constitutes a technical skill or hard skill. For this work, technical skills are defined as follows:

- Programming languages such as Python and R
- Areas of knowledge such as Machine Learning and Deep Learning
- Libraries and frameworks such as Pandas, NumPy, and Scikit-learn
- Applications or programs such as Excel, PowerPoint, and Jupyter Notebook
- Academic backgrounds such as Computer Science, Mathematics, Statistics, Physics, and Biology

The next step involved fine-tuning the models, where each was trained on the same labeled data for the skill recognition task. To train the models, the Google Colab platform was utilized in its professional version. Since two of the models feature transformer architectures, an integrated graphics processor (GPU) was required to accelerate the training process.

5. RESULTS AND DISCUSSION

When evaluating entity recognition problems, it is important to consider that words not classified by the algorithm, which are also not entities, will be counted as true negatives. This can create issues with evaluation metrics that rely on true negative counts, such as accuracy. Figure 10 illustrates the challenges associated with using accuracy in NER problems.

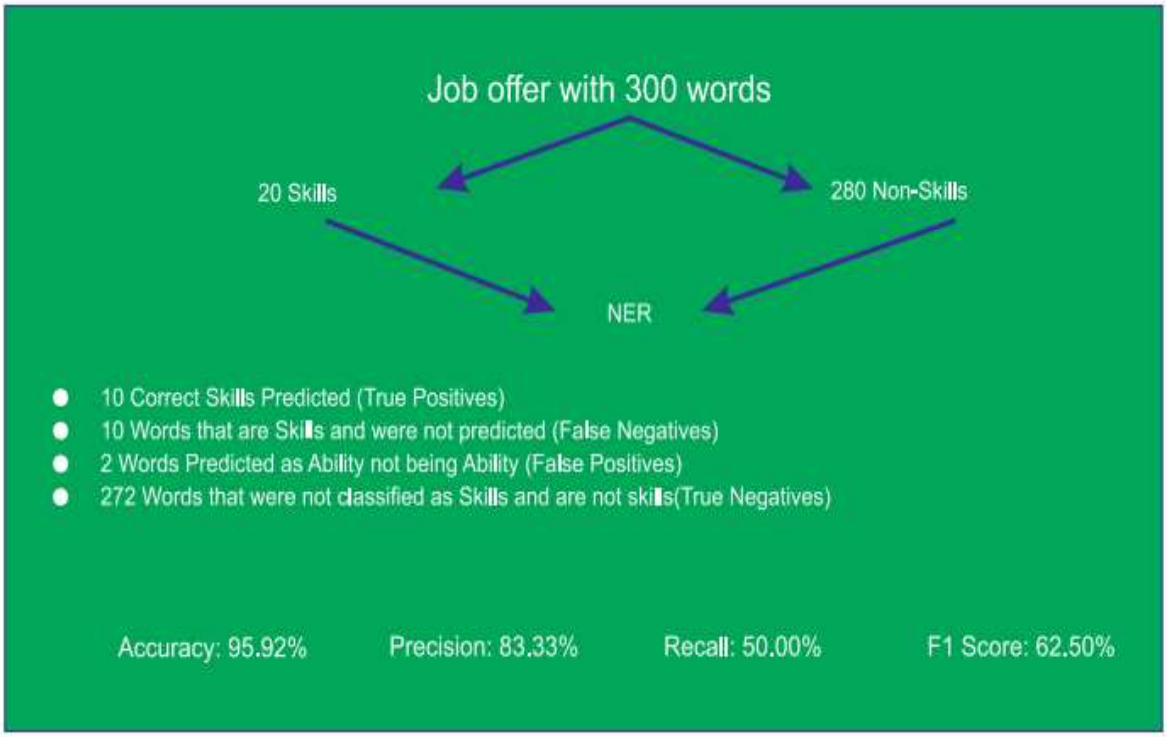


Figure 10. Example of Metrics

Three metrics were used to assess the model performance: Precision, Recall, and F1 Score.

To compute these metrics, the following results were analysed:

1. Words labeled as Skills that were also predicted by the algorithm as Skills (True Positives);
2. Words not labeled as Skills but predicted as Skills by the NER algorithm (False Positives);
3. Words labeled as Skills but not recognized as Skills by the NER algorithm (False Negatives).

Table 1 below presents the individual results of each of the three NER algorithms for each document tested in the NER column. Documents are numbered from 1 to 10, and the results of each algorithm for a given document can be found in the corresponding row.

Table 1 - Results of NER Algorithms

NER	Document	Total Abilities	Total-Predicted	Right-Prediction (True Positive)	Wrong-Prediction	Existing skill not found by algo (False Negatives)	Precision (%)	Recall	F1 Score
SPACY	1	19	24	24	0	5	100.00	82.76	90.57
SPACY	2	27	26	26	0	1	100.00	96.30	98.11
SPACY	3	11	10	9	1	2	90.00	81.82	85.71
SPACY	4	16	13	13	0	3	100.00	81.82	89.66
SPACY	5	5	5	5	0	0	100.00	100	100.00
SPACY	6	23	17	16	1	7	94.12	69.57	80.00
SPACY	7	12	10	8	2	4	80.00	66.67	72.73
SPACY	8	9	6	4	2	5	66.67	44.44	53.33
SPACY	9	24	26	24	2	0	92.31	100.00	96.00
SPACY	10	14	17	14	3	0	82.35	100.00	90.32
XLNet	1	29	21	19	2	10	90.48	65.52	76.00
XLNet	2	27	20	20	0	7	100.00	74.07	85.11
XLNet	3	11	9	9	0	2	100.00	81.82	90.00
XLNet	4	16	15	15	0	1	100.00	93.75	96.77
XLNet	5	5	4	4	0	1	100.00	80.00	88.89
XLNet	6	23	17	11	6	12	67.71	47.83	55.00
XLNet	7	12	5	5	0	7	100.00	41.67	58.82
XLNet	8	9	5	5	0	4	100.00	55.56	71.43
XLNet	9	24	17	16	1	8	94.12	66.67	78.05
XLNet	10	14	12	12	0	2	100.00	85.71	92.31
XLNet	1	29	23	27	3	2	90.00	93.10	91.53
XLNet	2	27	27	27	0	0	100.00	100.00	100.00
XLNet	3	11	10	10	0	1	100.00	90.91	95.24
XLNet	4	16	16	16	0	0	100.00	100.00	100.00
XLNet	5	5	5	5	0	0	100.00	100.00	100.00
XLNet	6	23	28	23	5	0	82.14	100.00	90.2
XLNet	7	12	10	10	0	2	100.00	83.33	90.91
XLNet	8	9	9	8	1	1	88.89	88.89	88.89
XLNet	9	24	26	24	2	0	92.31	100.00	96.00
XLNet	10	14	17	11	6	3	64.71	78.57	70.97

Figure 11 below presents a line graph illustrating the results of the Precision metric. It is evident that the XLNet algorithm outperformed the others, achieving 100 percent accuracy in 7 out of 10 documents and exceeding 90 percent accuracy in 9 of them. The BERT algorithm

demonstrated stable performance, slightly trailing behind XLNet in terms of accuracy, while the spaCy algorithm exhibited fluctuations from document to document.

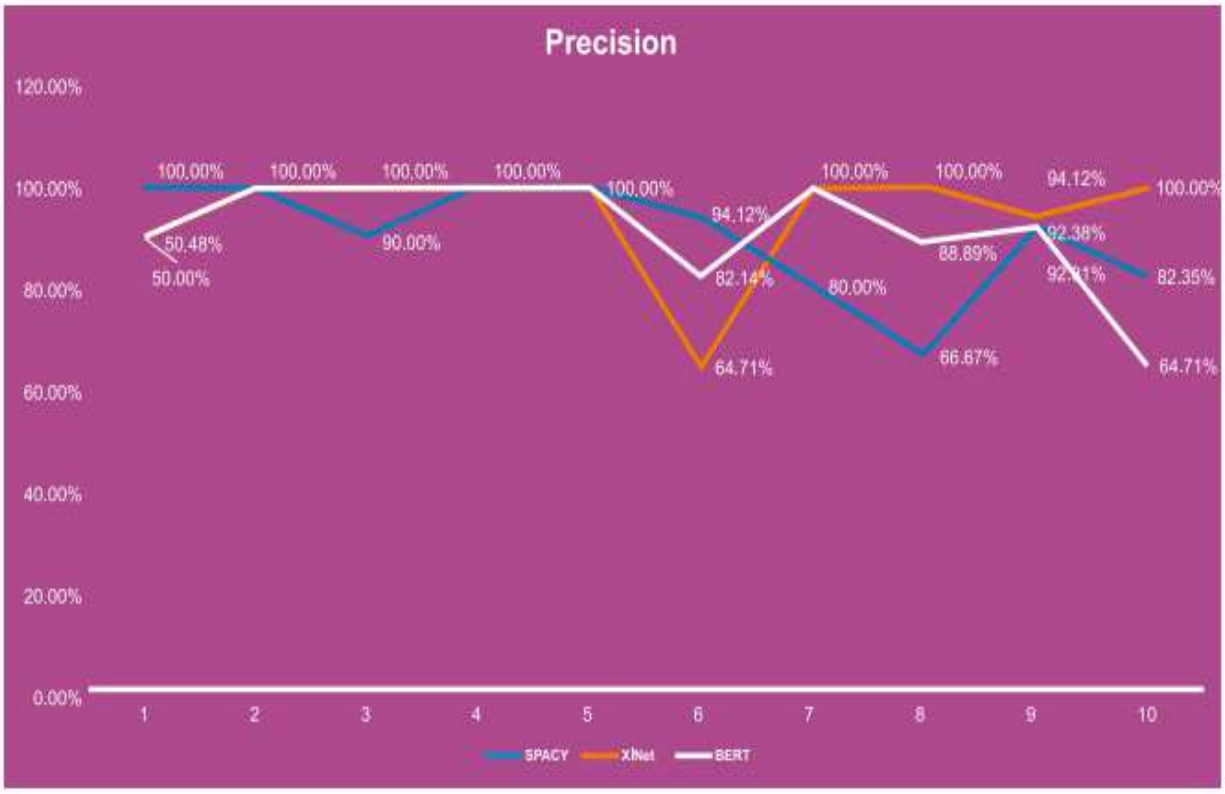


Figure 11. Precision

Regarding Recall, Figure 12 confirms that BERT was the most stable algorithm, demonstrating the best overall performance. It achieved results above ninety percent in 7 documents, 5 of which reached one hundred percent. In contrast, the other two algorithms performed significantly lower compared to BERT, with some documents showing values below fifty percent.

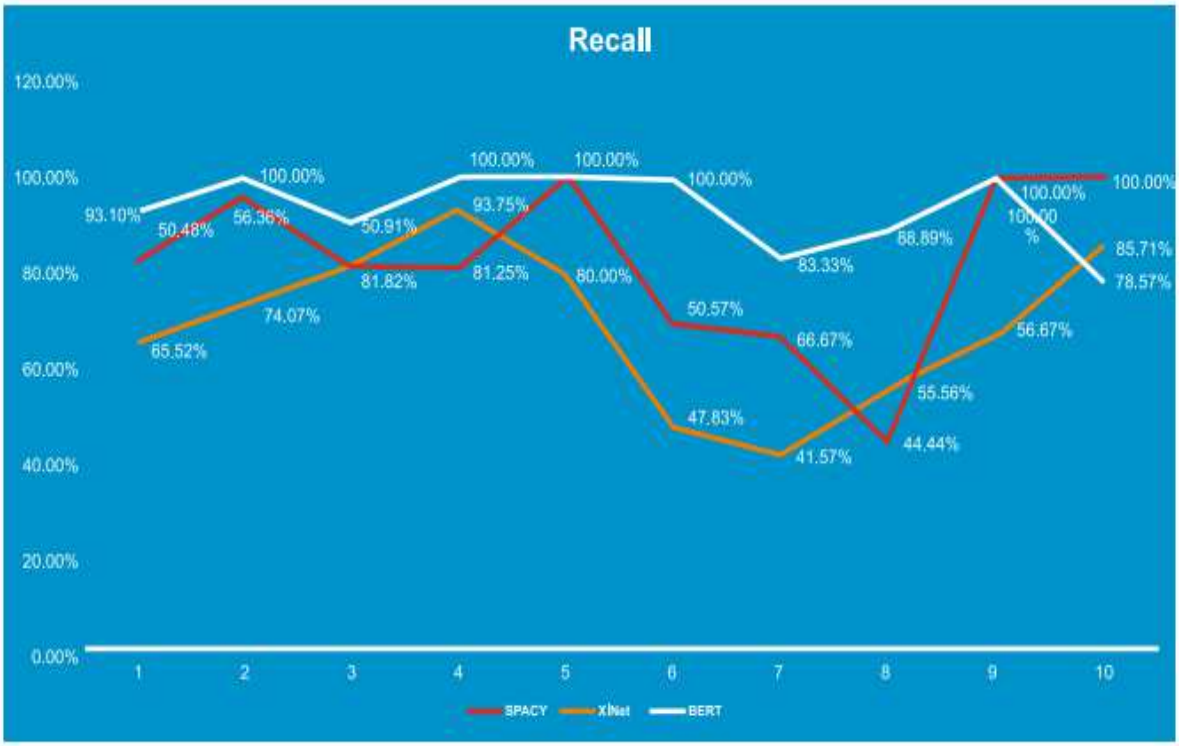


Figure 12. Recall

In the F1 Score, which aims to balance the results of Precision and Recall, BERT again excelled, achieving the best performance among the three algorithms in 9 of the 10 documents. In 8 of these documents, the F1 Score value exceeded ninety percent. Notably, the poor performance in Recall also impacted the F1 Score results for both XLNet and spaCy, as illustrated in Figure 13.



Figure 13. F1 Score

To determine the final model selection, the arithmetic mean of the results for each model was calculated across all documents for each metric. Table 2 below presents the final results, indicating that XLNet outperformed in terms of Precision, while BERT excelled in the other two metrics. Additionally, it is important to note that the difference in Precision between XLNet and BERT was 2.91%, while the difference in Recall between BERT and spaCy was 10.21%, and the difference in F1 Score between BERT and spaCy was 7.03%.

Table 2 - Final Result

NER	Precision	Recall	F1 Score
SPACY	91.26%	83.98%	86.14%
XLNet	95.22%	69.96%	80.19%
BERT	92.31%	94.19%	93.17%

These results indicate that the XLNet algorithm is highly reliable when recommending skills, achieving nearly 95% accuracy in avoiding incorrect recommendations. However, it also suggests that some skills present in the documents may go undetected by this algorithm. Therefore, the primary justification for using XLNet would be if the goal is to predict skills with the highest accuracy, regardless of the total number of skills identified. Figure 11 presents the results of the XLNet algorithm in a word cloud format.



Figure 14. WordCloud XLNet

spaCy demonstrated good performance in precision, with average results in recall. As a lightweight algorithm that is easy to train, it can be a suitable choice if hardware costs or training time are significant considerations for the project. Figure 15 illustrates the results of the spaCy algorithm.

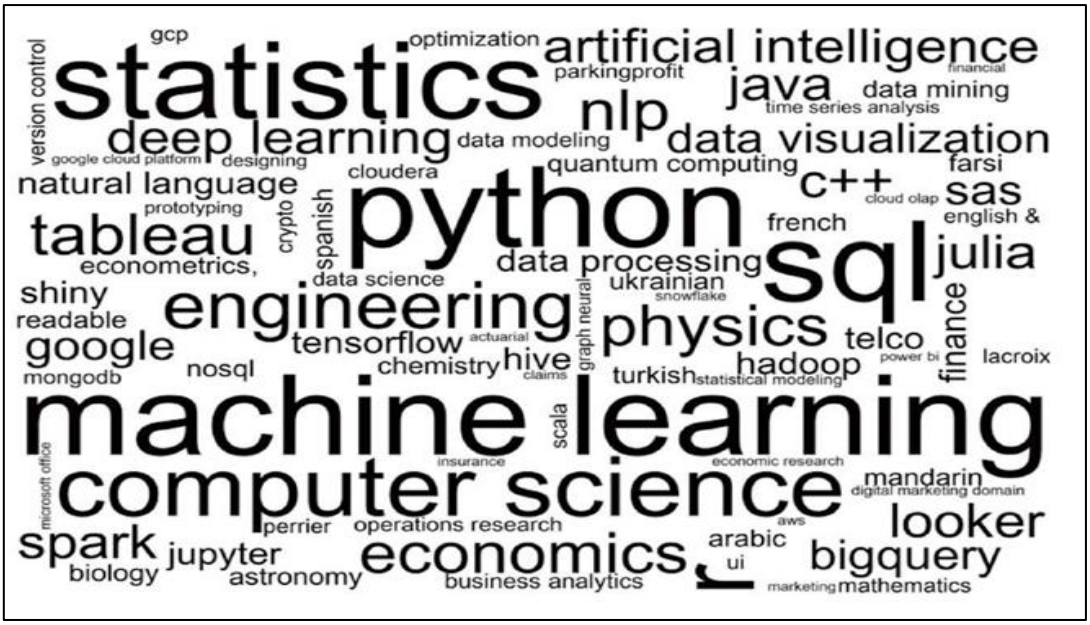


Figure 15. WordCloud Spacy

BERT emerged as a stable algorithm with outstanding performance across all metrics, making it the most reliable choice among the three algorithms. The fine-tuning of BERT demonstrated effectiveness, indicating that for the problem addressed in this work, the combination of the named entity recognition technique with transfer learning was a prudent decision. Figure 16 illustrates the results of the BERT algorithm.



Figure 16. WordCloud BERT

5. NOVELTY OF THE WORK

This research presents several novel aspects that enhance skill extraction models in the domain of data science job vacancies. The key contributions include:

5.1. Selection and Integration of Diverse Techniques

This work strategically integrates three distinct techniques TF-IDF, Word Embeddings, and Named Entity Recognition (NER) for skill extraction. The synergistic combination of these methods offers a comprehensive approach to addressing the complexities of job vacancy data.

5.2.2. Evaluation of State-of-the-Art Models

The study performs a comprehensive evaluation of three prominent pre-trained models SpaCy, BERT, and XLNet for Named Entity Recognition. This comparative analysis highlights their respective strengths and weaknesses, providing valuable insights into their suitability for skill extraction tasks.

5.2.3. Application of Transfer Learning

A notable contribution of this research is the application of Transfer Learning, which involves using pre-trained models on millions of data points for generalized tasks and fine-tuning them to identify skills in job vacancies. This technique significantly enhances the adaptability and performance of the models.

5.2.4. Detailed Performance Analysis

This research goes beyond mere model selection, offering a detailed performance analysis. Precision, recall, and F1 Score metrics are meticulously evaluated for each model, providing a nuanced understanding of their effectiveness in real-world skill extraction scenarios.

5.2.5. Practical Implications for Recruitment

The findings of this study have practical implications for talent recruitment, as the recommended model, BERT, demonstrates high precision and recall. The proposed methodology can enhance the efficiency of automated systems in identifying the technical skills required by the evolving job market.

6. CONCLUSIONS

- At the outset of this research, it became evident that monitoring the skills required for job vacancies poses a challenge due to the evolving job market. Consequently, it was crucial to develop an algorithm capable of detecting technical skills in job listings.
- In light of this, the research aimed to create a Natural Language Processing model for extracting technical skills in the field of data science. It is clear that this objective was

achieved, as the study effectively utilized NLP techniques and yielded significant results in the evaluation metrics.

- The first specific objective of this work was to analyze data processing techniques and evaluate their suitability for the problem at hand. Although stopword removal was not applied due to the chosen algorithm, this objective is considered achieved. The analysis of each processing technique was thoroughly examined, leading to the conclusion that not using this technique was the best choice.
- The second specific objective, which aimed to explore machine learning models to understand various techniques for extracting employment skills, was also achieved. This involved a thorough analysis of the literature on each supervised and unsupervised machine learning technique, as well as Natural Language Processing (NLP) and Transfer Learning. Ultimately, this analysis led to the selection of three potential candidates for skill extraction, with NER being chosen as the final model.
- The final specific objective was to apply the most promising techniques for skill extraction, which was successfully accomplished through the development of three models: XLNet, Spacy, and BERT. Additionally, it is noteworthy that the skill of Machine Learning, which was emphasized in the exploratory analysis as one of the most frequently mentioned skills, consistently appeared as the top skill identified by all three models used in this study.
- In conclusion, this study successfully extracted skills relevant to the field of data science.

6.1 LIMITATIONS AND RECOMMENDATIONS FOR FUTURE WORK

- The methodology for this study began with the selection of an initial area for skill extraction, followed by an in-depth literature review, the selection of extraction techniques, the creation of the model, and the presentation of results.
- Given the proposed methodology, this research focused on a specific employment area. One challenge in expanding this work to other fields is the need for knowledge about the skills relevant to each specific domain. A thorough understanding of the field is essential to ensure that technical skills are not overlooked during the labeling process.
- Additionally, the quantity of labeled data poses a limitation, as significant time is required to complete this task. Increasing the amount of labeled data could improve model performance and provide a larger dataset for evaluating the model's effectiveness.

- As a recommendation for future work, I suggest creating a new entity to identify interpersonal or soft skills, along with the labeling of these skills, which the model could then detect.
- Another important recommendation is to expand the model to other areas of work. One potential area to explore initially could be the broader technology sector.
- Lastly, I recommend investigating the sentence classification model proposed during the extraction techniques selection phase. Although this approach may not fully consider the semantics of the text, it has the potential to yield good results.

Disclosure statement

This manuscript is original and has not been published nor is it presently under consideration for publication elsewhere. The author states that there are no conflicts of interest.

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Data availability statement

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

REFERENCES

1. Skondras, P., Zervas, P., & Tzimas, G. (2023). Generating Synthetic Resume Data with Large Language Models for Enhanced Job Description Classification. Future Internet, 15(11), 363.
2. Fang, C., Qin, C., Zhang, Q., Yao, K., Zhang, J., Zhu, H., & Xiong, H. (2023, August). Recruitpro: A pre-trained language model with skill-aware prompt learning for intelligent recruitment. In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (pp. 3991-4002).
3. Stojmenović, M. (2022). Understanding Job Requirements Using Natural Language Processing. In Sinteza 2022-International Scientific Conference on Information Technology and Data Related Research (pp. 457-463). Singidunum University.
4. Vanetik, N., & Kogan, G. (2023). Job Vacancy Ranking with Sentence Embeddings, Keywords, and Named Entities. Information, 14(8), 468.
5. C. Zhu, H. Zhu, H. Xiong et al., "Person-job fit: adapting the right talent for the right job with joint representation learning," *ACM Transactions on Management Information Systems*, vol. 9, no. 3, pp. 12.1–12.17, 2018.
6. C. Qin, H. Zhu, T. Xu et al., "An enhanced neural network approach to person-job fit in talent recruitment," *ACM Transactions on Information Systems*, vol. 38, no. 2, pp. 1–33, 2020.
7. C. Zhu, H. Zhu, H. Xiong, and P. Ding, "Recruitment market trend analysis with sequential latent variable models," in *Proceedings of the 22nd ACM SIGKDD International Conference*, pp. 383–392, San Francisco, CA,USA, 2017.
8. Paul, S., & Lal, K. (2021). Technology Intensity and Employment in the Indian Economy. Arthaniti: Journal of Economic Theory and Practice, 20(1), 34-52. https://doi.org/10.1177/0976747919895326
9. Pinzone, M., Fantini, P., Perini, S., Garavaglia, S., Taisch, M. & Miragliotta, G. (2017). Jobs and Skills in Industry 4.0: An Exploratory Research. IFIP International Conference on Advances in Production Management Systems (APMS), Hamburg, Germany. (pp.282-288)
10. Zanine, N., Dhawan, V, (2015, January). Text Mining: An introduction to theory and some applications (pp. 38-39). Research Matters: A Cambridge Assessment publication.
11. Liddy, E.D. (2001). Natural Language Processing. In Encyclopedia of Library and Information Science, 2nd Ed. NY. Marcel Decker, Inc.

12. Bachtiar, F. A., Pradana, F., & Yudiari, R. D. (2019). Employee Recruitment Recommendation Using Profile Matching and Naïve Bayes. *2019 International Conference on Sustainable Information Engineering and Technology (SIET).*, pp. 94-99. Retrieved from https://ieeexplore.ieee.org/document/8985988
13. Korkmaz, O. (2019). Personnel selection method based on topsis multi-criteria decision making method. *Uluslararası İktisadi ve İdari İncelemeler Dergisi*, pp. 284-287. doi:https://doi.org/10.18092/ulikidince.468486
14. Rabcan, J., Vaclavkova, M., & Blasko, R. (2017). Selection of appropriate candidates for a type position using C4.5 decision tree. *International Conference on Information and Digital Technologies (IDT)*, pp. 332-338. doi:https://doi.org/10.1109/dt.2017.8024318
15. Fernández-García, A. (2019). The fight against discrimination in access to employment through the anonymous curriculum vitae. IUSLabor. Journal of Labor Law analysis, (1), 233-266.
16. Cardenas, C., & Hernandez, J. (2020). "Hire what is similar to you…" Discourse and discrimination in the recruitment and selection processes of personnel in large companies in Chile. Discourse & Society, (4), 790-822. http://www.dissoc.org/ediciones/v14n04/DS14%284%29Cardenas&Hernandez.html
17. Hmoud, B., & Laszlo V. (2019). Will artificial intelligence take over human resources recruitment and selection? *Network Intelligence Studies, VII*(13 (1/2019)), 21–30.
18. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaise, L., & Polosukhin, I. (2017). Attention is all you need. In Advances in neural information processing systems (pp. 5998- 6008).
19. Wolf T., Debut L., Sanh V., Chaumond J., Delangue C., Moi A., Cistac P., Rault T., Louf R., Funtowicz M., Davison J., Shleifer S., Platen P. V., Ma C., Jernite Y., Plu J., Xu C., Scao T. L., Gugger S., Drame M., Lhoest Q., & Rush A. (2020, October). Transformers: State-of-the-art natural language processing. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (pp. 38-45).
20. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
21. Chaima Antit , Seifeddine Mechti, and Rim Faiz, TunRoBERTa: A Tunisian Robustly Optimized BERT Approach Model for Sentiment Analysis, Advances in Intelligent Systems Research, volume 175, pp. 227-231, 2022.
22. Shuyong Wei, Defa Yu and Chenguo Lv, *J. Phys.: Conf. Ser.* **1693** 012076
23. Floridi, L., & Chiriatti, M. (2020). GPT-3: Its nature, scope, limits, and consequences.

Minds and Machines, *30*(4), 681-694.

24. C. Qin, H. Zhu, T. Xu, C. Zhu, and L. Jiang, “Enhancing person-job fit for talent recruitment: an ability-aware neural network approach,” in *Proceedings of the 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, pp. 25–34, Ann Arbor, MI, USA, 2018.